

Student Attrition Machine Learning

Implementation traceability | Prompts, assumptions, feature exclusions, and design decisions

Authoritative implementation: Machine Learning Model.ipynb (Databricks export 2026-08-16)

Controlling SDD: constitution, specification, plan (amendments A-1 to A-6), tasks, and phase prompts

Prepared: 20 August 2026

1. Current approved artefact

Item	Approved value
Target	<code>is_twelve_month_student_attrition</code>
Output table	<code>workspace.student_aggregate.student_attrition_risk</code>
Features	21 source features (5 numeric, 16 categorical)
Split	Deterministic 70/15/15 by hashing <code>student_deidentified_hash</code> (A-4)
Classifier	Spark ML <code>RandomForestClassifier</code>
RF settings	100 trees, <code>maxDepth</code> 7, <code>minInstancesPerNode</code> 20, <code>featureSubsetStrategy</code> <code>sqrt</code> , <code>seed</code> 42 (A-5)
Preprocessing	Training-derived medians and levels; stateless Spark expressions; manual one-hot with unseen/NULL bucket; <code>VectorAssembler</code> as Transformer only (A-6)
Primary metric / threshold	Positive-class recall; 0.50 on unrounded probability

2. Model assumptions and known limitations

- 1.Synthetic, deidentified data only. No claim is made about real student records.
- 2.Census-date snapshot: features must be available at prediction time.
- 3.Student-level outcome, not course-level leaving.
- 4.One eligible fact row per student in this generation (973,770 = 973,770).
- 5.Positive class is rare (~5.51%); balanced training weights are required.

6. Stored percentage is a ranking score, not a calibrated probability (L-3).

7. Serverless constraints are binding: no cache/persist; 100 MB per Spark ML model; 1 GB session cache.

ID	Limitation
L-1	Synthetic target field of education is decoupled from the course-dimension FoE the model can see.
L-2	All 21 features are close to flat against the target. Low discrimination is the correct result for this generator.
L-3	Weighted probabilities are not calibrated risk percentages. The application must present <code>attrition_risk_percentage</code> as relative risk, not as a percentage chance of leaving.

3. Approved features (21) and exclusions

Canonical ownership: fact table owns demographics, load, and cumulative credit points; teaching-period dimension owns teaching period; course dimension owns course group and field of education. A-1 added `course_admission_load_category` and `commencing_continuing_period`.

Numeric	Categorical
<code>age_at_census</code> ; <code>eftsl</code> ; <code>enrolment_year</code> ; cumulative credit points enrolled / passed / failed / withdrawn	<code>socioeconomic_status</code> ; <code>regional_remote_status</code> ; <code>student_gender</code> ; <code>student_is_international_student</code> ; <code>student_is_first_nations_student</code> ; <code>attendance_mode</code> ; <code>commencing_continuing</code> ; <code>commencing_continuing_period</code> ; <code>course_admission_load_category</code> ; <code>course_group</code> ; broad / narrow / detailed primary field of education; <code>teaching_period</code>

Excluded (validation corrections, not competing experiments): `target` and `is_twelve_month_course_attrition`; TCSI/withdrawn attrition flags; outcome hashes; raw keys; `age_band`; duplicate international and commencing encodings; zero-variance `is_enrolment` / `is_study_load`; census dates; `course_level`; `calendar_year`; study-area and unit field of education (A-2); offering, organisation, module, and thesis attributes with no usable populated columns; 38 further all-NULL fact columns.

4. Amendment history

ID	Decision
A-1	Two extra usable fact attributes; 19 to 21 features.
A-2	Canonical ownership over exhaustive duplicate columns.
A-3	Synthetic key width 4 to 12 to fix lpad collisions.
A-4	Hash split instead of <code>randomSplit</code> (serverless cache ban).
A-5	RF 300/10/5 reduced to 100/7/20 for Serverless model-size caps. This is a computational constraint, not a literature-driven hyperparameter search.
A-6	Stateless preprocessing instead of fitted Spark ML Pipeline. Statistical contract remains train-only transforms; mechanism changed for Spark Connect graph size.

5. Controlling prompts (Development Contract)

Master prompt: PySpark/Spark ML; student attrition target; exclude leakage and identifiers; RF 100/7/20/sqrt/42; training-derived stateless preprocessing; VectorAssembler as Transformer only; direct RF fit; 70/15/15; recall primary; threshold 0.50; persist only the approved Delta table; three human-gated phases; no model-comparison experiments.

Phase 1: fact-centred STAR assembly from the enrolment EFTSL fact table; profile 12 tables; one row per student or stop; join only verified 1:1 keys; feature manifest with include/exclude reasons.

Phase 2: consume the Phase 1 manifest unchanged; hash split; train-only medians and manual one-hot; class weights from training; fit RF directly; log forest plus preprocessing contract; evaluate validation then untouched test; do not write the prediction table.

Phase 3: reuse the training-derived contract and fitted forest; do not refit; persist six columns after T27; retrieve by `student_deidentified_hash`.

7. Out of scope for ML Model development

Course-attrition model; alternative-model benchmarking; competing feature subsets; automated feature selection; broad hyperparameter search; automated intervention; generative-AI advisor briefing; MLflow promotion; scheduled retraining; real-record validation without industry-partner access.